

Python Built-in Functions with Examples

This document covers some of the most important Python built-in functions with clear descriptions and examples. These functions are ready to use without any import.

1. print()

Description: Displays output on the screen.

```
print("Hello, Python!")
```

```
print("Sum of 5 and 3 is", 5 + 3)
```

Using sep and end

```
print("Python", "is", "fun", sep="-") # Python-is-fun
```

```
print("Hello", end=" "); print("World") # Hello World
```

f-string formatting

```
name = "Reshma"
```

```
score = 95.6789
```

```
print(f"Hello {name}, your score is {score:.2f}")
```

Printing lists

```
data = [1, 2, 3]
```

```
print("List:", data)
```

Escape characters

```
print("Hello /n World")
```

Writing to a file

```
with open("output.txt", "w") as f:  
    print("This goes into a file.", file=f)
```

2. len()

Description: Returns the number of items in a string, list, tuple, etc.

```
name = "Python"  
print(len(name)) # 6  
  
print(len([1, 2, 3, 4])) # 4
```

3. max() and min()

Description: - max() → returns the largest value. - min() → returns the smallest value.

```
numbers = [3, 7, 1, 9, 4]  
print(max(numbers)) # 9  
print(min(numbers)) # 1  
  
words = ["apple", "banana", "cherry"]  
print(max(words)) # cherry (alphabetically last)  
print(min(words)) # apple
```

4. round()

Description: Rounds a number to nearest integer or to given decimal places.

```
print(round(5.67))    # 6
print(round(5.67, 1)) # 5.7
```

5. abs()

Description: Returns the absolute (positive) value of a number.

```
print(abs(-10)) # 10
print(abs(3.5)) # 3.5
```

6. type()

Description: Returns the type (class) of a value or variable.

```
print(type(5))      # <class 'int'>
print(type("Hello")) # <class 'str'>
print(type([1,2,3])) # <class 'list'>
```

7. input()

Description: Takes input from the user as a string.

```
name = input("Enter your name: ")
print("Hello", name)

# Converting input to integer
age = int(input("Enter your age: "))
print("You will be", age + 1, "next year.")
```

8. sum()

Description: Adds up all elements in an iterable.

```
numbers = [10, 20, 30, 40]
print(sum(numbers))      # 100
print(sum(numbers, 50))  # 150 (adds 50 to the sum)
```

9. sorted()

Description: Returns a new sorted list.

```
nums = [5, 2, 9, 1]
print(sorted(nums))      # [1, 2, 5, 9]
print(sorted(nums, reverse=True)) # [9, 5, 2, 1]
```

```
names = ["Alice", "Bob", "Charlie"]
print(sorted(names))     # ['Alice', 'Bob', 'Charlie']
```

10. id()

Description: Returns the unique identity (memory address) of an object.

```
x = 100
y = 100
print(id(x))
print(id(y)) # same id (small integers are cached)
```

```
z = [1, 2, 3]
print(id(z))
```

11. help()

Description: Shows documentation for an object.

```
help(len)
```

```
help(print)
```

```
import math
```

```
help(math)
```